(almost)

Everything I Know About Plant Genome Editing

Problems, Peculiarities and Practical Tips

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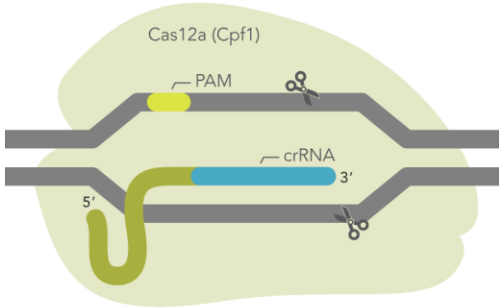
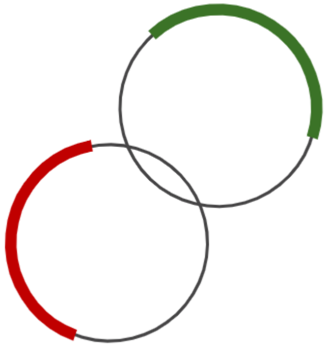
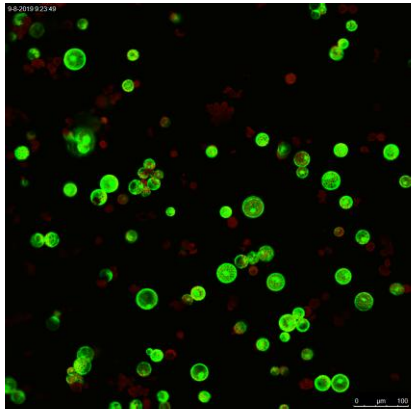
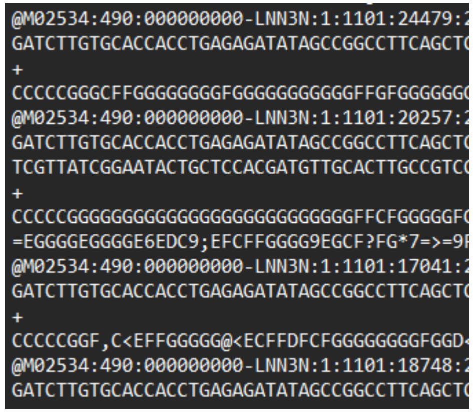
What *do* I know about plant genome editing?

- MSc Biotechnology
  - MSc thesis “Expanding the CRISPR toolbox in tomato”
  - MSc internship at a company to set up genome editing pipeline
- PhD thesis “CRISPR specificity in tomato”
- Postdoc “Directed evolution for improved Cas12a base editing”









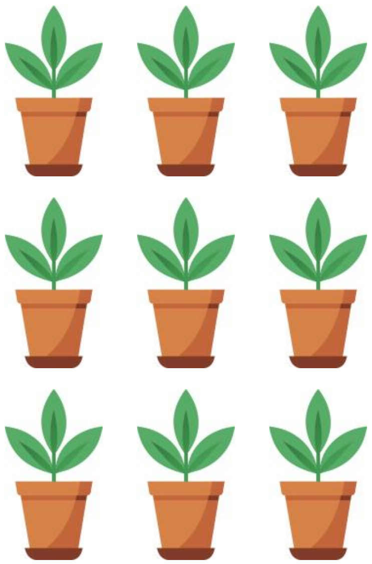
1. Challenges in Plant Genome Editing
2. Plant Genome Editing Pipeline

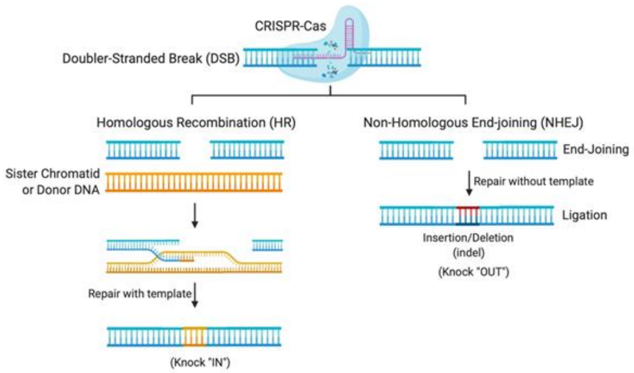
Plants are pretty cool.

CO_2 in, O_2 out
Food
Useful compounds
Pretty

Plants have some peculiarities.

Challenges in plant genome editing






```
>plantgenome
AGTGGCGTGTAAAG
GTGTCAGTAGGTGT
CCCTGCACATGATG
CGCGATGTGCCGGT
ATCGTGACGTTCAT
CGCATTTACCCGGG
TCACGTGCAGTCTA
GTCGCTATATGGAG
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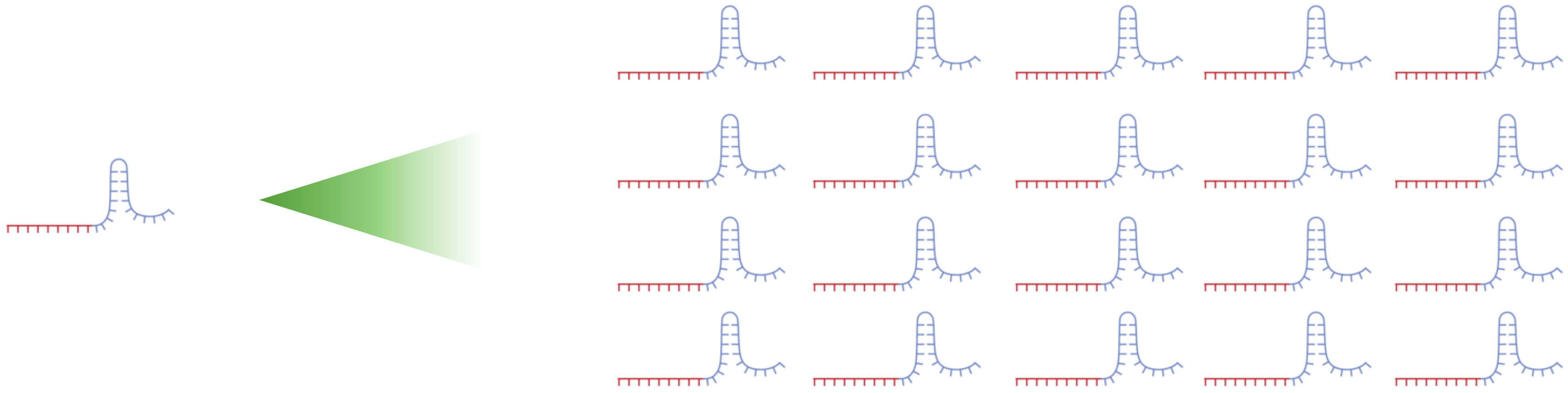
Scale

Editing Outcomes

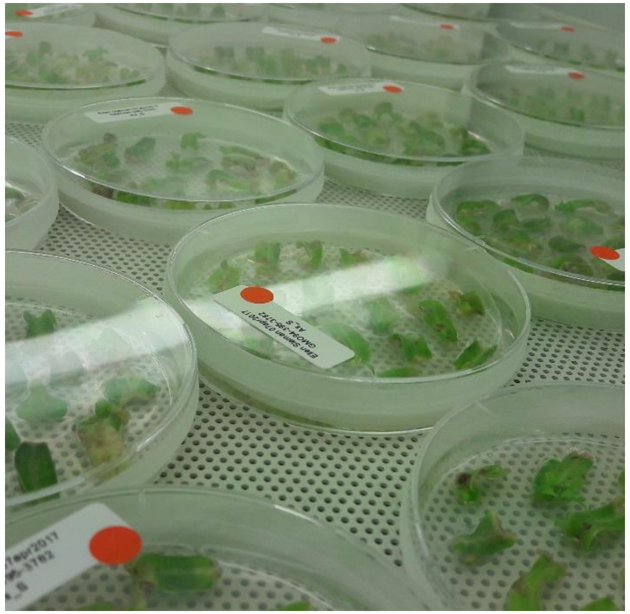
Transformation protocols & genetic resources

What do I mean by scale?

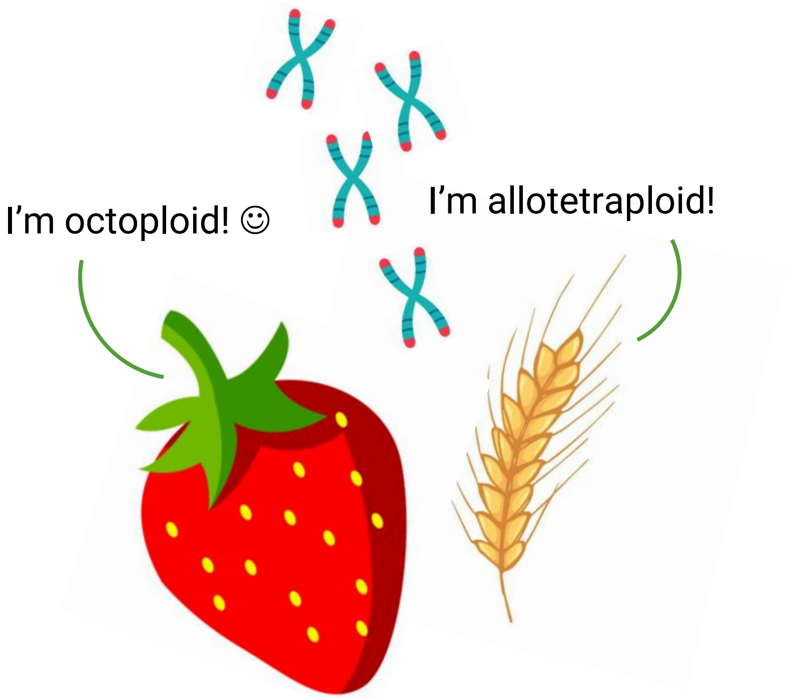
- The amount of transformed plants you generate
- The amount of genes you target or gRNAs you use



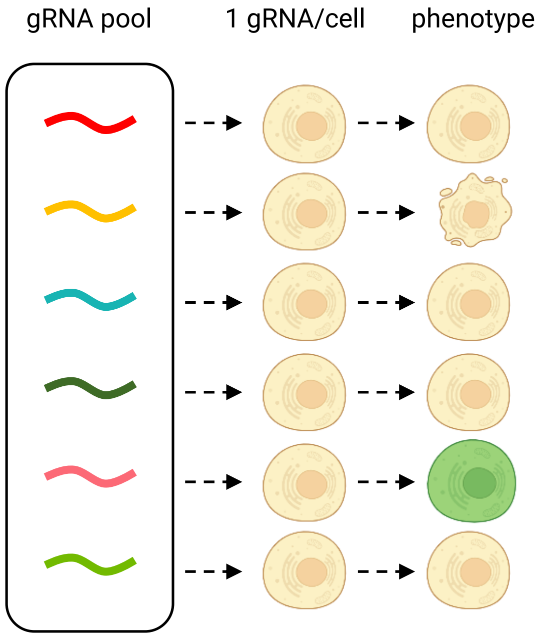
Why are we limited in scale?



Transformation procedure and generation time



Polyploidy, complex and repetitive genomes

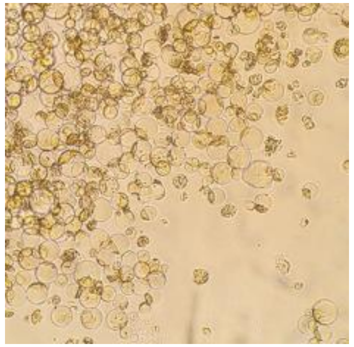


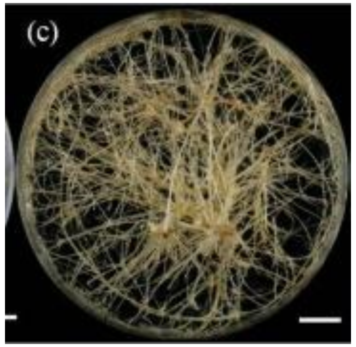
No great way to do pooled screens

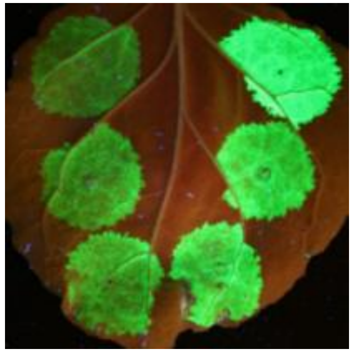
Origin and evolution of the octoploid strawberry genome https://www.nature.com/articles/s41588-019-0356-4

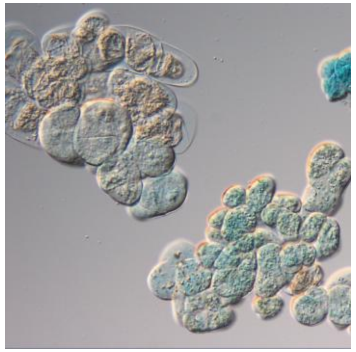
What are some ways around this?

- Transient systems
  - Protoplast assays
  - Hairy root transformation
  - *Nicotiana benthamiana* leaf infiltrations
  - Cell suspension cultures
  - …









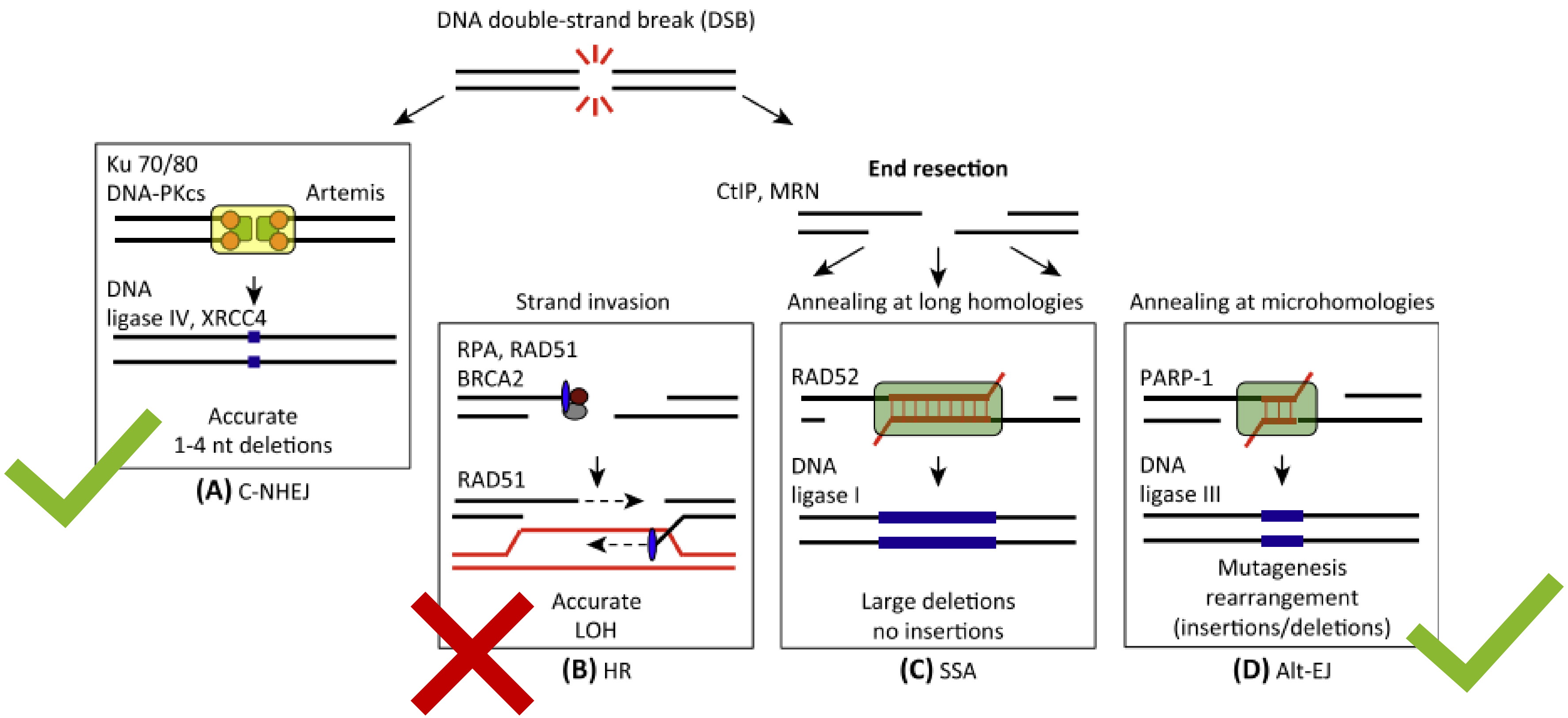
- However: all have limitations
  - For example, there's no plant development
  - How relevant those limitations are depends on your research question

The sad consequences

- All kinds of elegant CRISPR applications are out of reach for plants (for now)
  - MAGESTIC
  - Genome-wide knockout, CRISPRa or CRISPRi screens
  - Directed evolution *in planta*
  - Doesn’t mean people aren’t trying!

Reviews on CRISPR screens in plants:
Gaillochet et al., 2021, Plant Cell
Pan et al., 2023, Current Opinion in Biotechnology
But see also, for example:
Arndell et al., 2024, Nature Plants – Pooled screens in protoplasts
Berman et al., 2025, Nature Communications – gRNA library screen in tomato

Editing outcomes



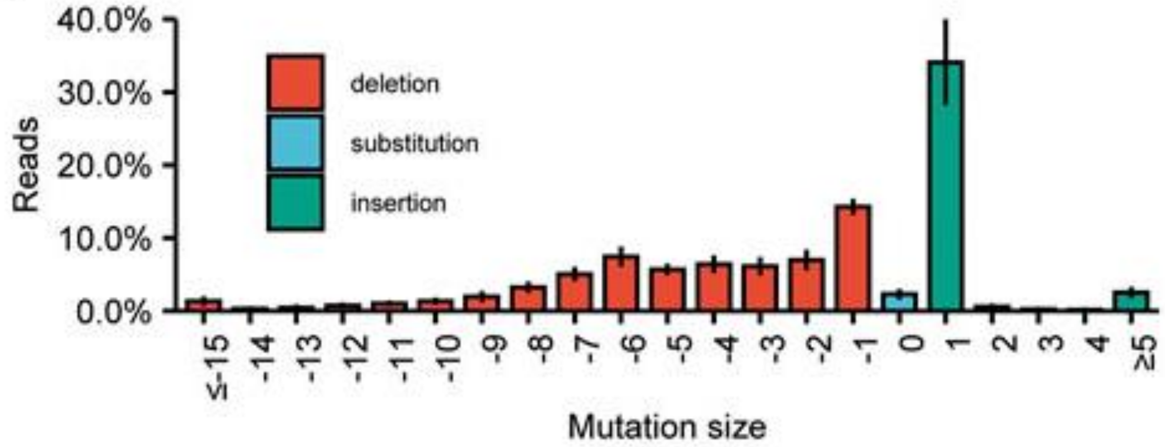
Breaking things is easy.
Precision engineering is difficult.

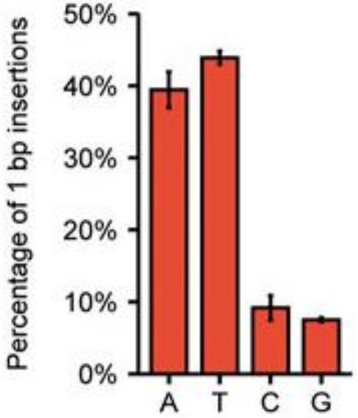
Ceccaldi et al., 2016, Trends in Cell Biology

What are some common editing outcomes?

Based on 89 **Cas9** gRNAs in tomato protoplasts

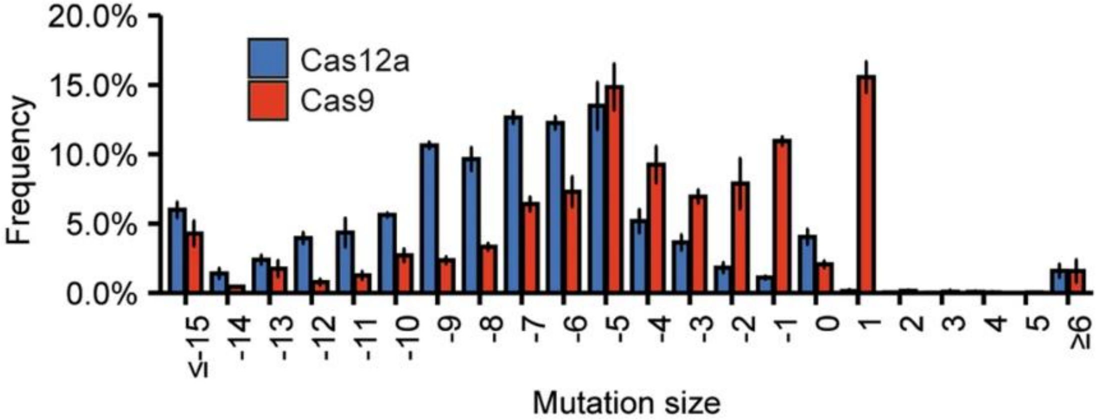
A





Based on 35 overlapping **Cas9 and Cas12a** gRNAs in tomato protoplasts

e



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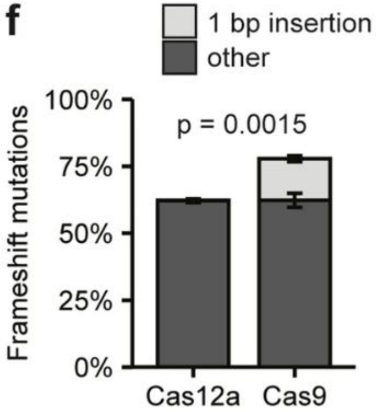


Slaman et al., 2023, Frontiers in Genome Editing
Slaman et al., 2024, Scientific Reports

Are these different from mammalian cells?

Cas9 introduces 1 bp 5’ overhangs, resulting in duplication of the -4 nt

gLYS2-1^{C}

5’ GAACCTGTG TTGTGGAGA
3’ CTTGGACACA ACACCTCT

5’ overhang
↓

5’ GAACCTGTGT TTGTGGAGA
3’ CTTGGACACA AACACCTCT

fill-in
↓

5’ GAACCTGTGTTTGTGGAGA
3’ CTTGGACACAAACACCTCT

+1 bp insertion

Base insert at predicted site	gLYS2-1^{C}
+A	2
+T	74
+G	24
+C	-

gLYS2-1^{W}

5’ GAACCTGT GTTGTGGAGA
3’ CTTGGACAC AACACCTCA

5’ overhang
↓

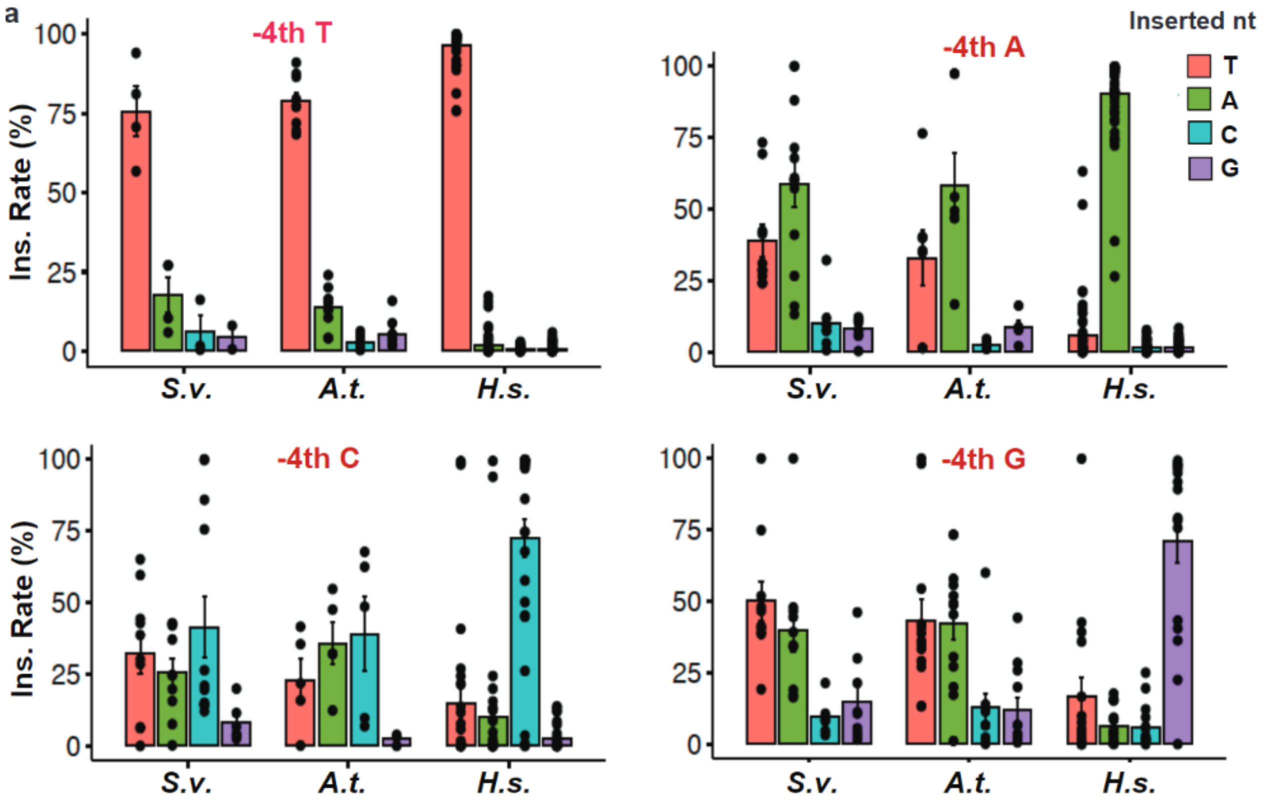
5’ GAACCTGTG GTTGTGGAGA
3’ CTTGGACAC CAACACCTCA

fill-in
↓

5’ GAACCTGTGGTTGTGGAGA
3’ CTTGGACACCAACACCTCA

+1 bp insertion

Base insert at predicted site	gLYS2-1^{W}
+A	-
+T	-
+G	100
+C	-



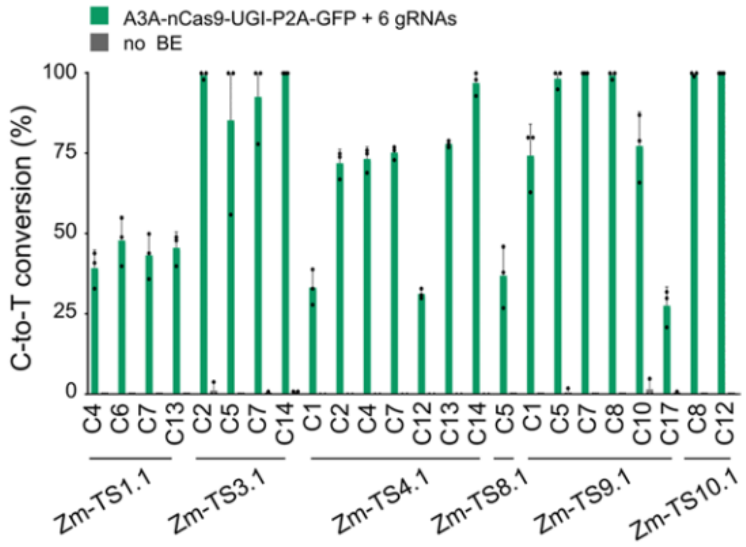
Yes, they are!

Lemos et al., 2018, PNAS

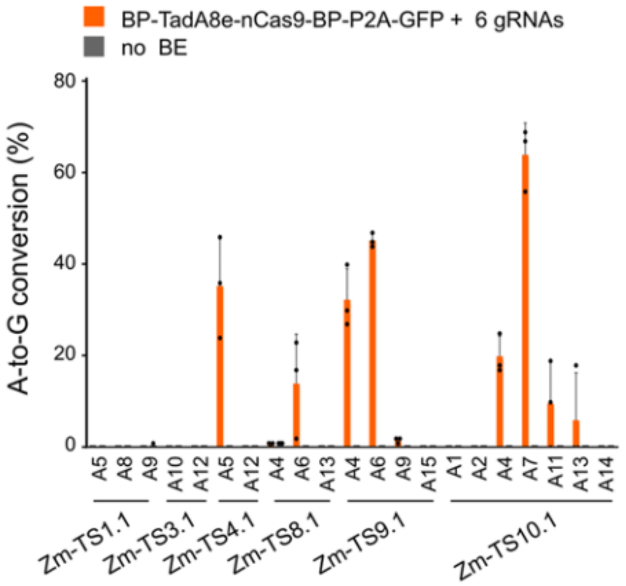
Weiss et al., 2024, Nature Communications

What if I don't want to make a knockout?

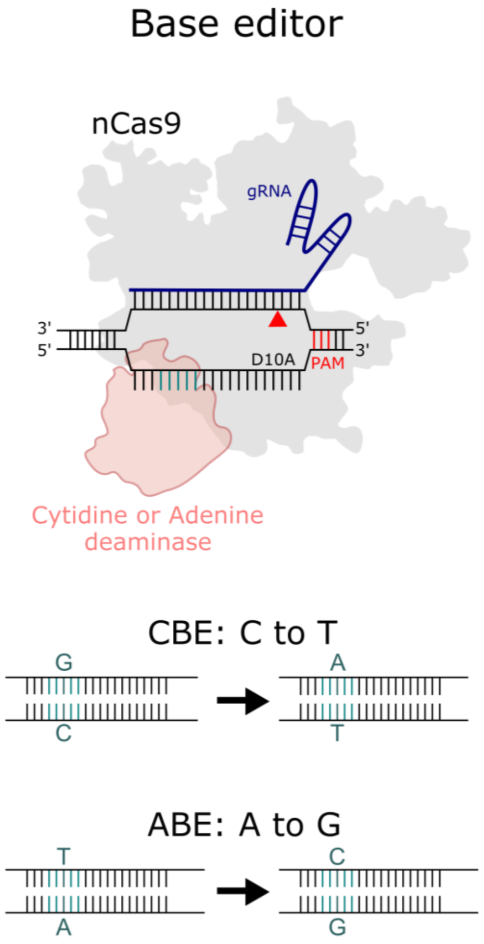
- Base editing has decent efficiencies
- Prime editing may be feasible if you budget time for optimization and trial-and-error



CBE in maize protoplasts



ABE in maize protoplasts



Gaillochet et al., 2023, Genome Biology

Transformation protocols & genetic resources

- With no way to get transformants, you're going to spend your PhD figuring out a way to get transformants
- It's difficult to design gRNAs if you don't know where they're supposed to fit
- Not such a big deal for the majority of people working on well-characterized model organisms, but a big deal for more applied research that's done, for example, in Africa

Plant Genome Editing Pipeline

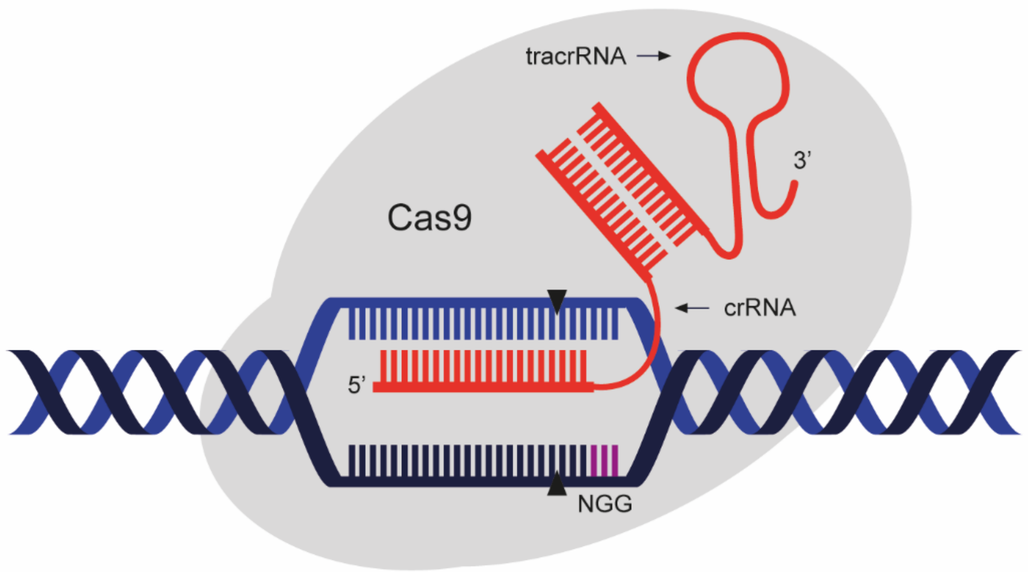
Design | Clone | Deliver | Genotype | … | Phenotype

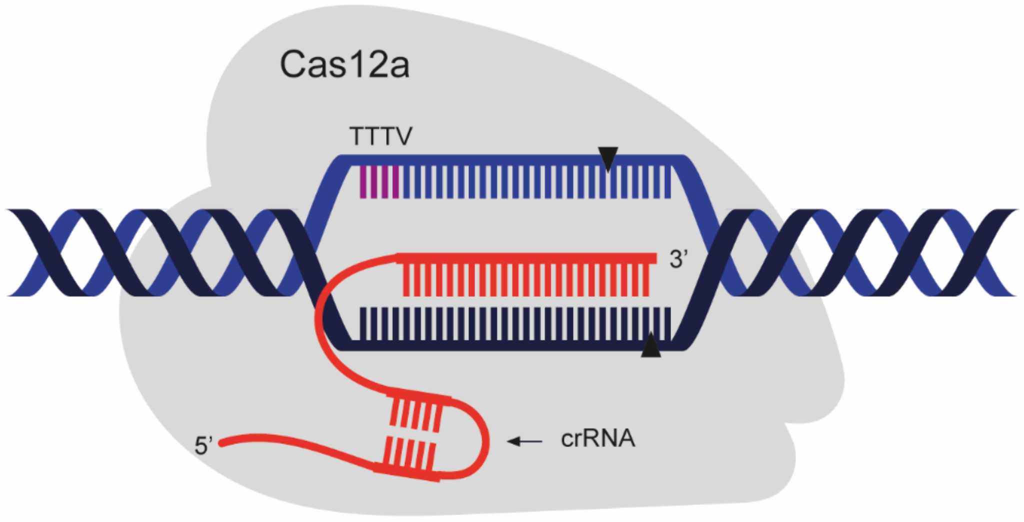
(Grow | Genotype) * *n*

Experimental design

- ★ Tip: Start at the end
- What is the desired outcome of your experiment?
  - “I want to test if a knockout of gene X increases drought tolerance”
  - “I want to mess with this promoter and see the effect on expression”
- What do you need to get that data?
  - “Stably transformed plants with knockouts of gene X, with CRISPR construct segregated out”
  - “Population of cells with all kinds of different mutations in this promoter”
- What is the desired outcome of your experiment on the DNA level?
  - Knockout?
  - Specific mutation?

What Cas-effector am I going to use?



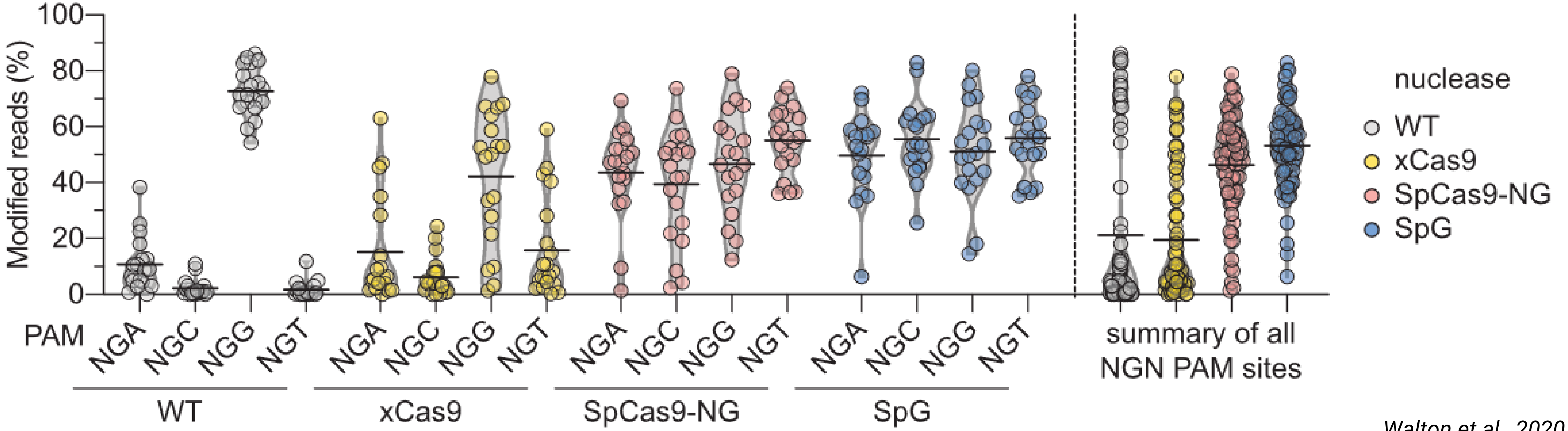


NGG PAM
The go-to enzyme
More out-of-frame mutations

TTTV PAM (promoter regions)
Reliable, but less data
Larger deletions

What Cas-effector am I going to use?

- Various PAM-relaxed variants, such as Cas9-SpRY
- Various high-specificity variants
- ★ If you can, avoid these variants. The efficiency is often lower than the wildtype.



Walton et al., 2020, Science

gRNA design

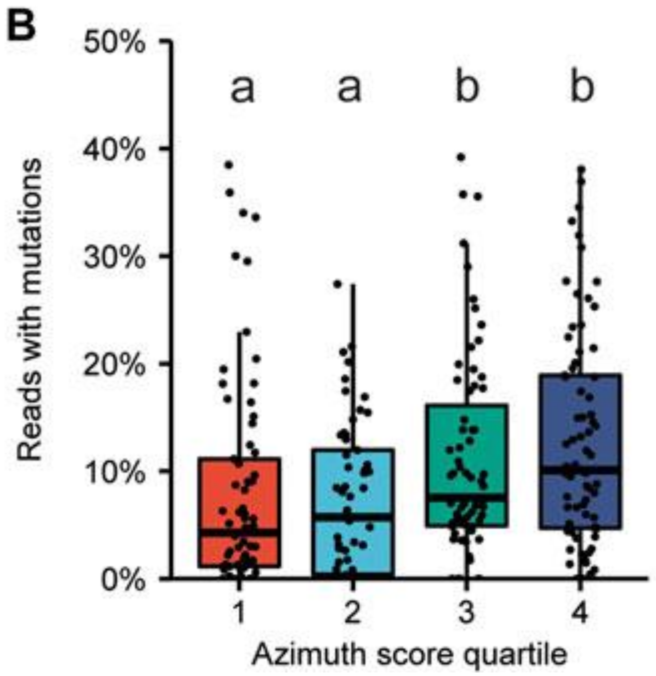
- Are you sure this is the sequence you're targeting?
  - Are you sure you're sure?
- Can you genotype this sequence?
  - If not, why bother targeting it?
  - Plant promoters, introns and UTRs are notoriously difficult to amplify
- Tip: do a trial PCR and send it for Sanger

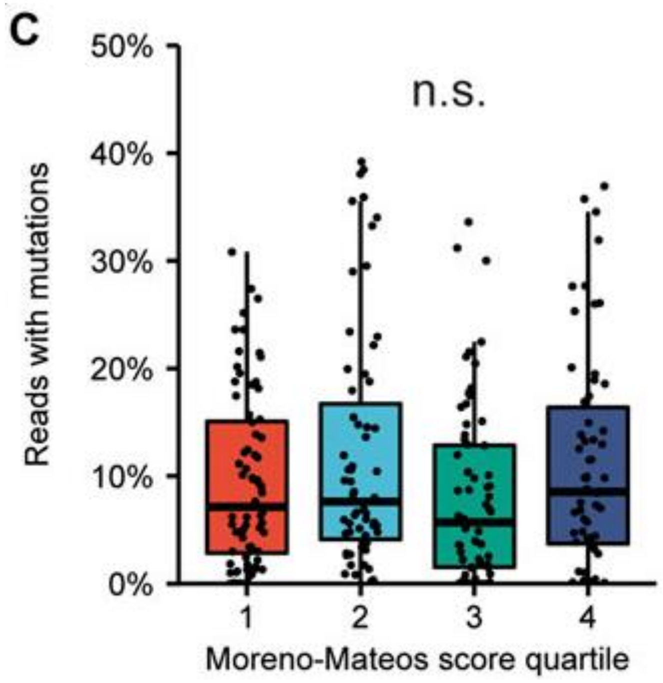
What are some tools to design gRNAs?

- CRISPR-P http://cbi.hzau.edu.cn/cgi-bin/CRISPR2
- CRISPOR https://crispor.gi.ucsc.edu/crispor.py
- CHOPCHOP https://chopchop.cbu.uib.no/
- … and many more
- ★ Poly-T stretches (≥4 T's) function as termination signals for the PolIII promoters we use to express gRNAs: avoid those

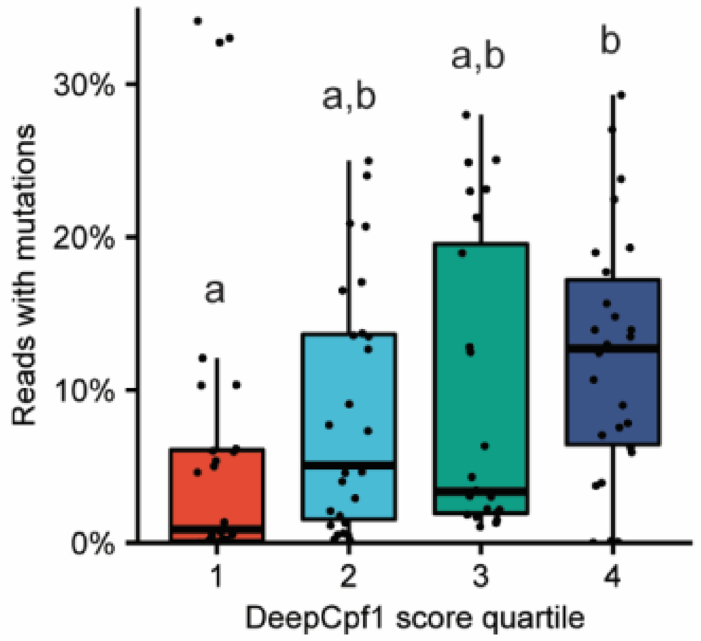
How reliable are efficiency predictions?

Cas9





Cas12

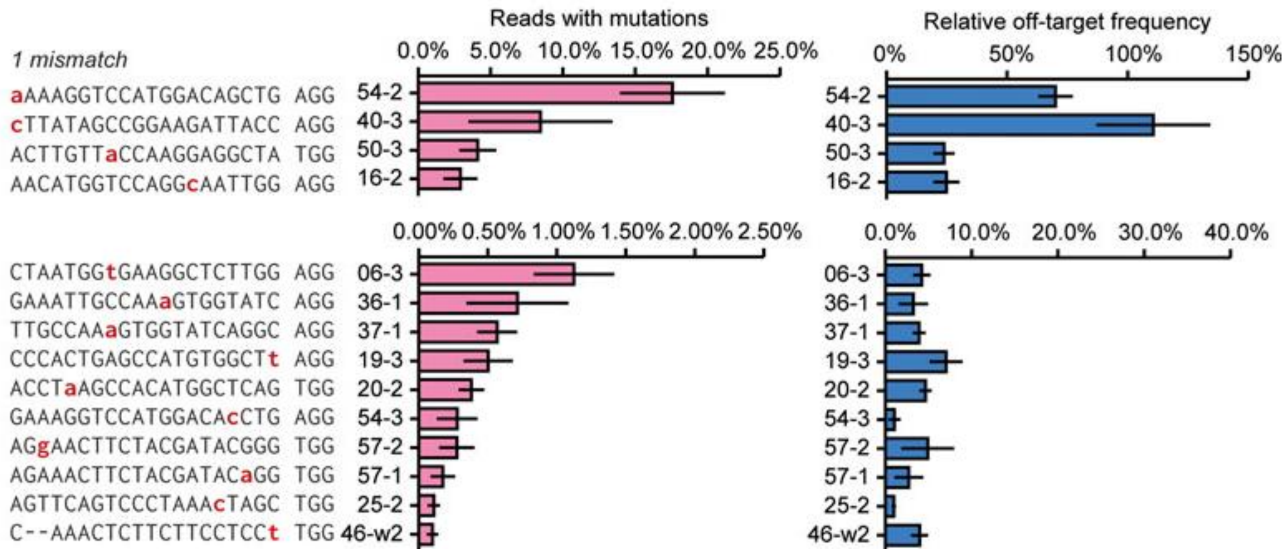


Not very.

Slaman et al., 2023, Frontiers in Genome Editing
Slaman et al., 2024, Scientific Reports

How worried should I be about off-targets?

- If there are ≥ 3 mismatches, don’t worry
- If there are 2 mismatches, you will generally be able to find a mutant that only has a mutation at the target
- If there is 1 mismatch, it depends on the location



Slaman et al., 2023, Frontiers in Genome Editing

Off-targets can be a feature instead of a bug

Multigene Knockout Utilizing Off-Target Mutations of the CRISPR/Cas9 System in Rice

Masaki Endo[1], Masafumi Mikami[1,2] and Seiichi Toki[1,2,*]

```
CDKB2 (Chr.8)    AAGGTCGGGGAGGGGACGTACGGG
CDKA1 (Chr.3)    AAGATTGGGGAGGGCACGTACGGG
CDKA2 (Chr.2)    AAGATCGGGGAGGGGACGTACGGG
CDKB1 (Chr.1)    AAGGTGGGGGAAGGGACGTACGGG
```

B

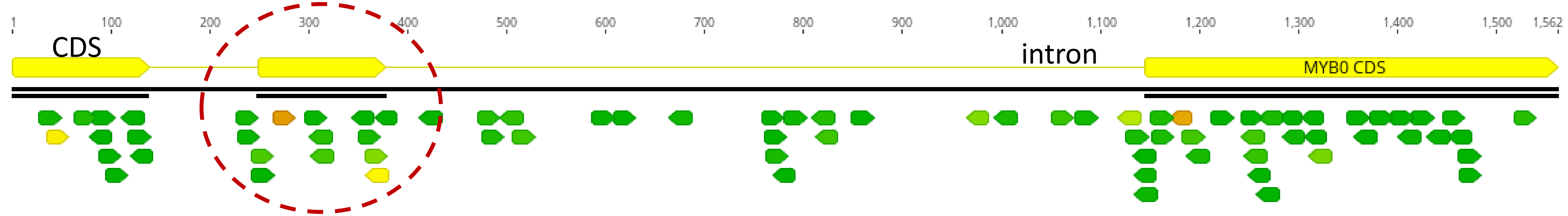
```
CDKB2
CTGGAGAAGGTCGGGGAGGGGACGTACGGGAAGGTGTACAAGGCGCGG    WT
CTGGAGAAGGTCGGGGAGGGGA--TACGGGAAGGTGTACAAGGCGCGG    (-2) #17-4
CTGGAGAAGGTCGGGGAGGGGAC-TACGGGAAGGTGTACAAGGCGCGG    (-1) #17-2,6,8
CTGGAGAAGGTCGGGGAGGGGACGATACGGGAAGGTGTACAAGGCGCGG   (+1) #17-9,10,11,12,13

CDKA2
GTGGAGAAGATCGGGGAGGGGACGTACGGGGTGGTGTACAAGGGCAAGC   WT
GTGGAGAAGATCGGGGAGGGGACGTTACGGGGTGGTGTACAAGGGCAAGC  (+1) #17-4
GTGGAGAAGATCGGGGAGGGGAGGTACGGGGTGGTGTACAAGGGCAAGC   (m1) #17-2,8,13
GTGGAGAAGATCGGGGAGGGGAC-TACGGGGTGGTGTACAAGGGCAAGC   (-1) #17-9(homo)
GTGGAGAAGATCGGGGAGGGGA-GTACGGGGTGGTGTACAAGGGCAAGC   (-1) #17-12

CDKB1
CTGGAGAAGGTGGGGGAAGGGACGTACGGGAAGGTGTACAAGGCG       WT
CTGGAGAAGGTGGGGGAAGGGAC-TACGGGAAGGTGTACAAGGCG       (-1) #17-2
CTGGAGAAGGTGGGGGAAGGGACGATACGGGAAGGTGTACAAGGCG      (+1) #17-7
CTGGAGAAGGTGGGGGAAGGGACGTTACGGGAAGGTGTACAAGGCG      (+1) #17-4,5,6,8,11,12,13
CTGGAGAAGGTGGGGGAAGGG--------------GTACAAGGCG       (-14)#17-5
```

Design tips for knockouts

- Try to target essential gene regions (if known)
- Use multiple gRNAs
  - 3 is usually feasible and compatible with cloning systems
- Target roughly the middle third of a gene
  - Too close to the start and an alternative start codon might take over
  - Too close to the end and your protein might be somewhat functional
- Select an easy-to-genotype region



What promoter, nuclear localization signal and terminator should I use?

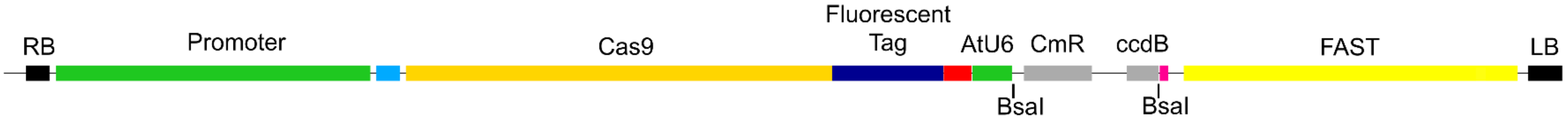
- Dependent on your experimental system
  - Monocot (U3) or dicot (U6)?
  - Floral dip (embryo-active) or another type of transformation (strong, constitutive)?

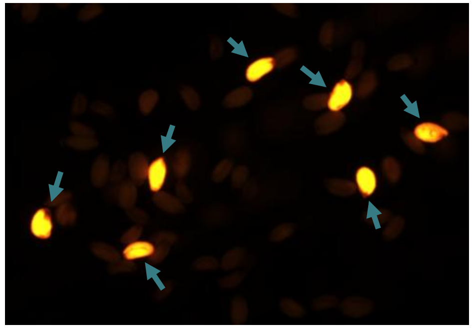
Species	Cas promoter	Cas terminator	NLS	gRNA promoter
Arabidopsis	RPS5A	T7	BP	AtU6
Zea mays	ZmUbi	G7T	BP	TaU3
Triticum aestivum	ZmUbi	G7T	BP	TaU3
Solanum lycopersicum	2x35S	NOS	2xSV40	U6_26

Expression system in Arabidopsis: Develtere et al., 2024, The Plant Journal
Cas12a in Arabidopsis: Xin et al., 2025, Plant Physiology

Clone

- You’ve selected your guides, your nuclease… now what?
- ★ Sprinkle in some fluorescence as a little treat to make your life easier (& more colourful)
  - Allows positive selection in T_0, negative selection in T_1





FAST positive Arabidopsis seeds

Cloning

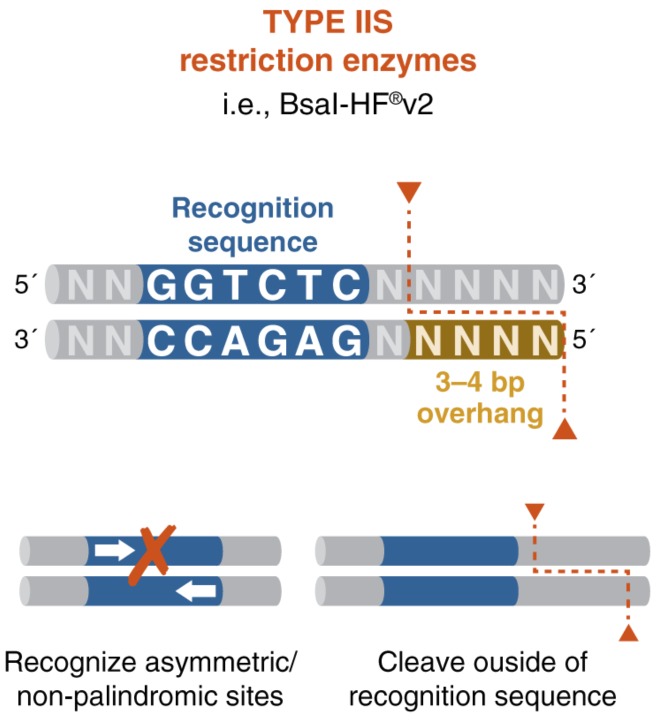
- There are multiple modular cloning systems available for plant researchers
  - Cloning is unlikely to be a bottleneck
  - Most* cloning systems are based on Golden Gate cloning and/or Gibson assembly
  - My personal favourites: MoClo and MultiGreen
- Tip: whatever you use, make sure everything works *in silico*
- … and avoid Gateway cloning

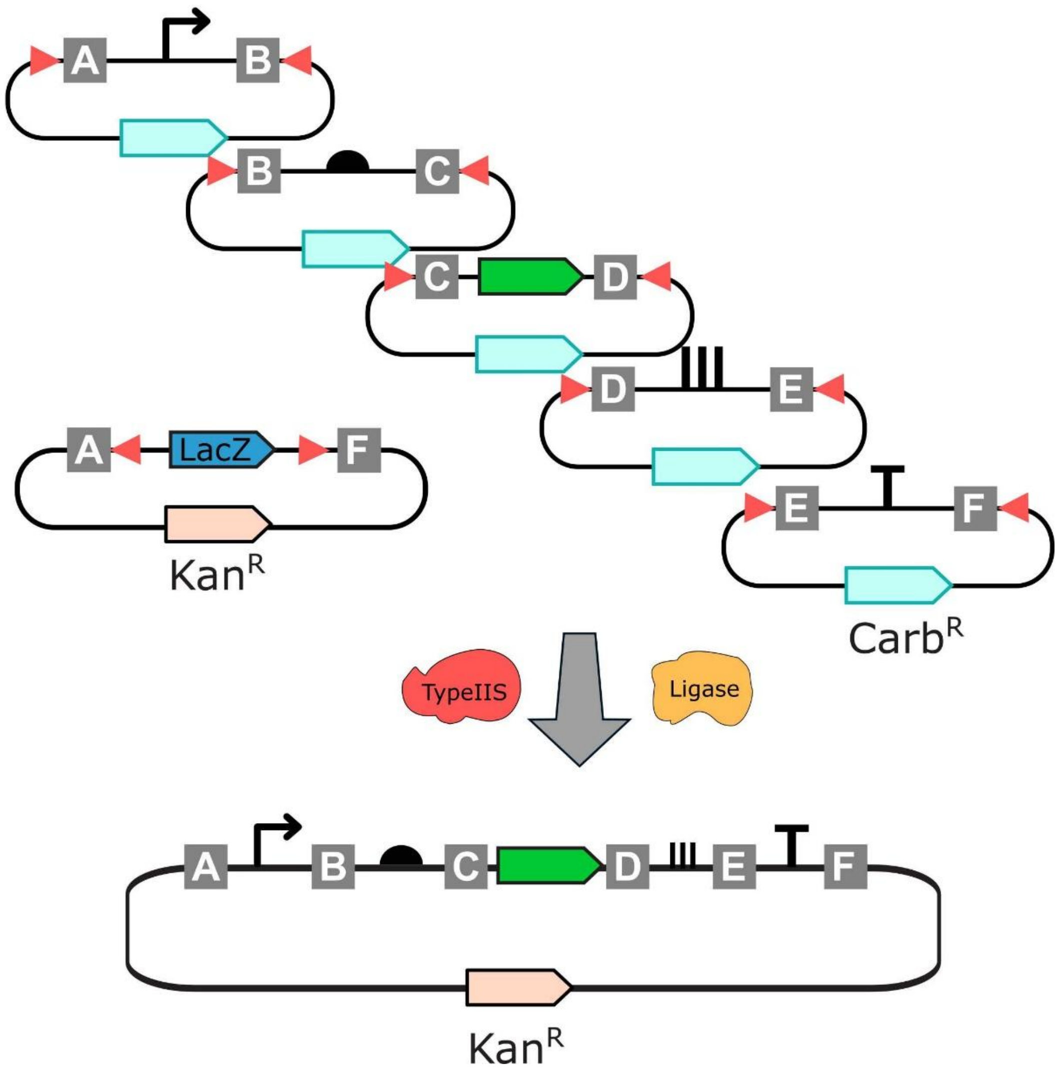
free

Benchling geneious prime SnapGene by Dotmatics

* Read: The best

Golden Gate Cloning



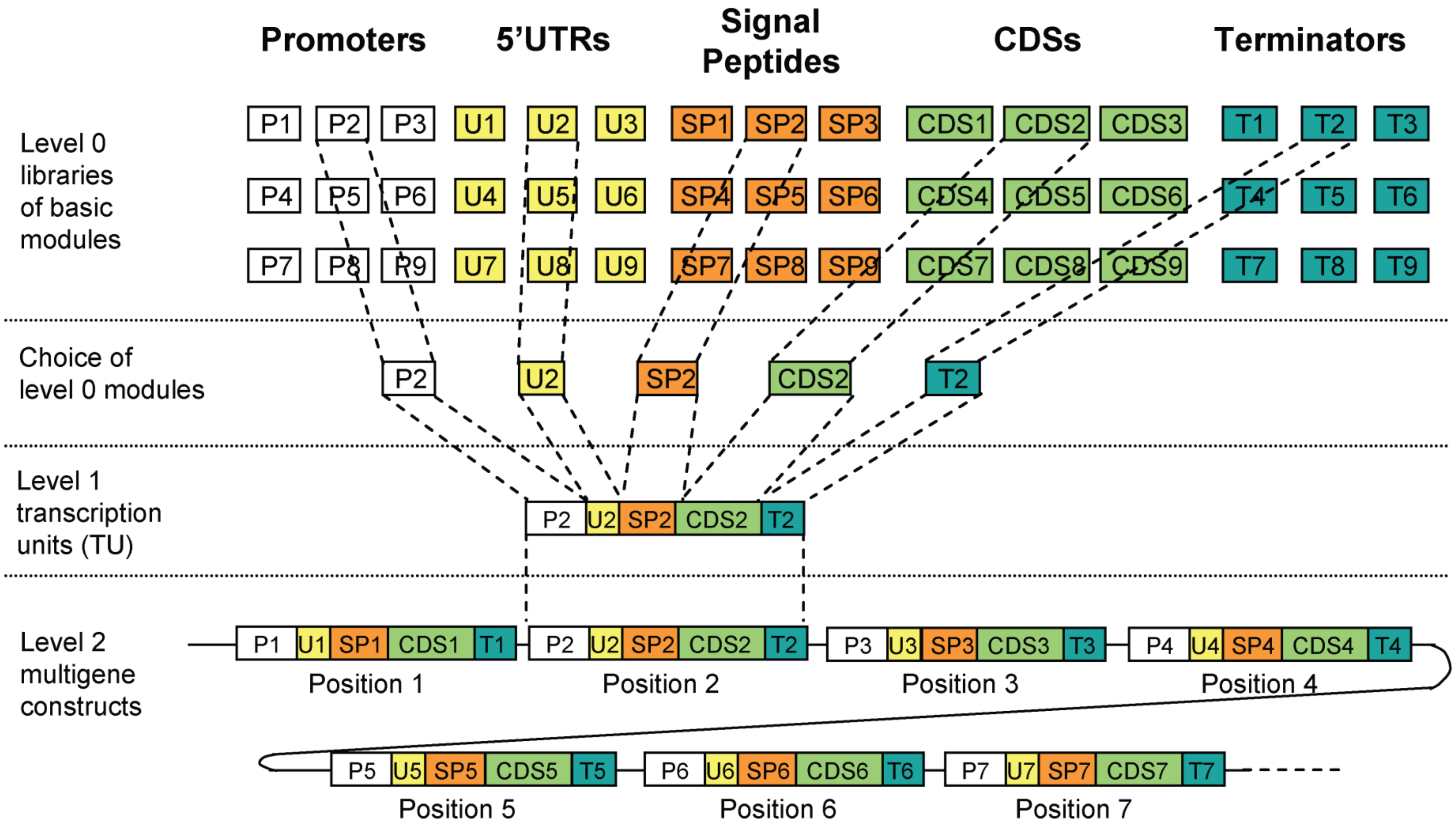


Alternating cycles of digestion (37°C) and ligation (16°C)
Reaction can only proceed in one direction: very efficient

"You can spit in the tube and it will be fine"

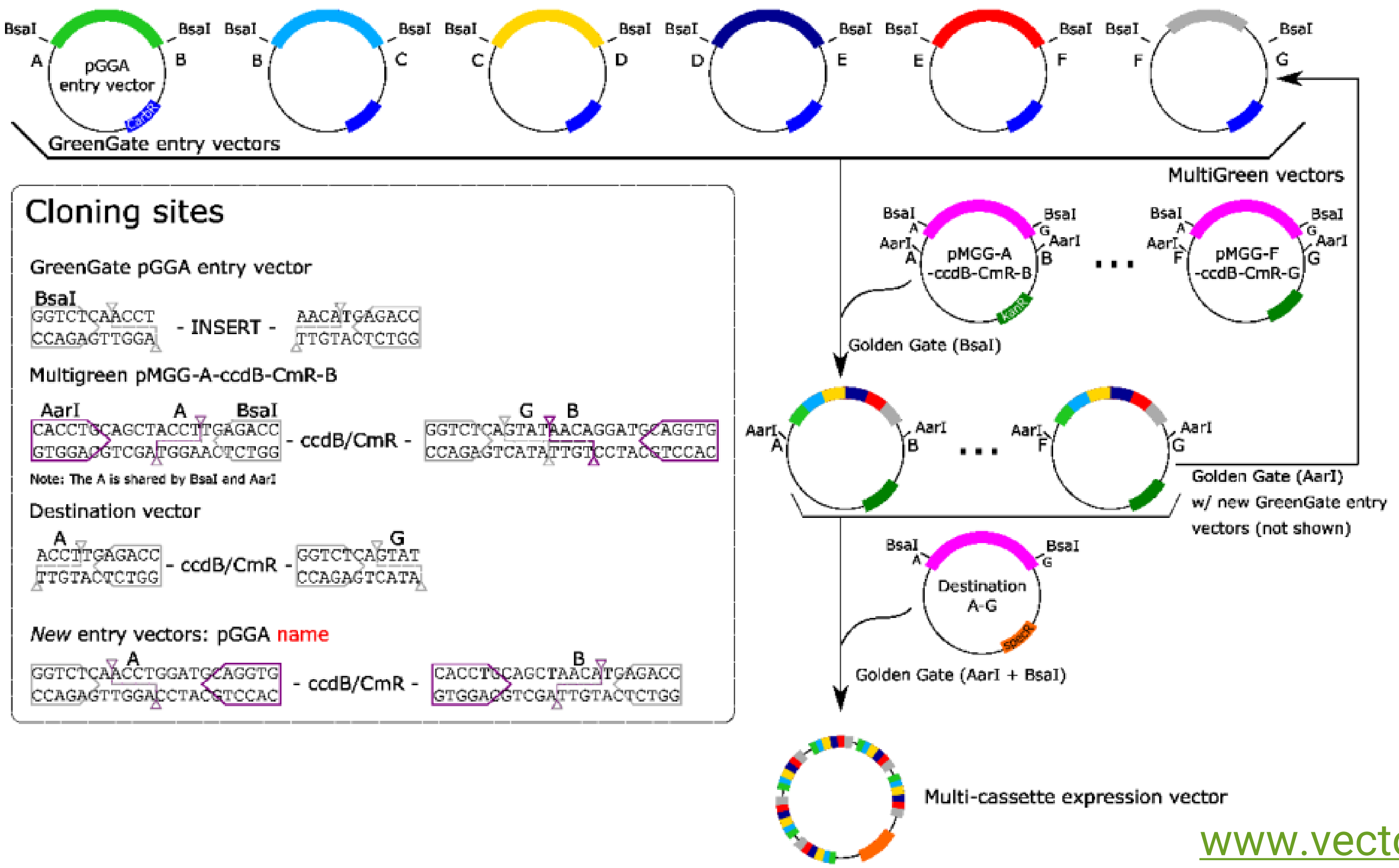
- Anonymous Scientist at a Maize Conference, 2024

MoClo (Modular Cloning)



Basic kit: Weber et al., 2011, Plos One
Plant Parts: Engler et al., 2014, ACS Synthetic Biology

MultiGreen



www.vectorvault.vib.be
https://vectorvault.vib.be/CRISPR

GreenGate: Lampropoulos et al., 2013, PLOS ONE

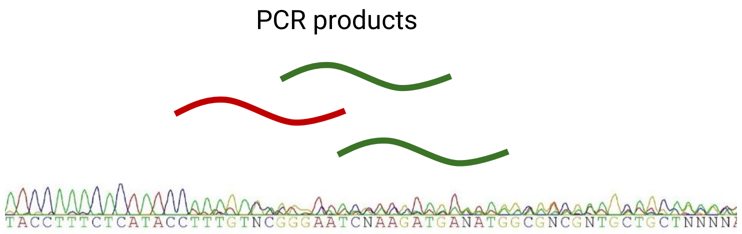
Deliver

- Agrobacterium
- Protoplast transfection
- Virus
- Biolistics (the famous particle bombardment aka gene gun)

Genotyping

- Genotyping T_0's can be tricky
  - Mix of alleles
  - Tissue culture may sometimes result in increased ploidy
  - Chimeric plants (>2 alleles): which alleles are in the germline?
  - Large deletions may disrupt primer binding sites: "fake homozygotes"
- Fluorescence can help reduce the workload
  - Select fluorescent T_0's to genotype, as they have CRISPR
  - Select non-fluorescent T_1's to genotype, as they don't have CRISPR
  - Sometimes, fluorescence is silenced! Confirm with PCR.

Genotyping

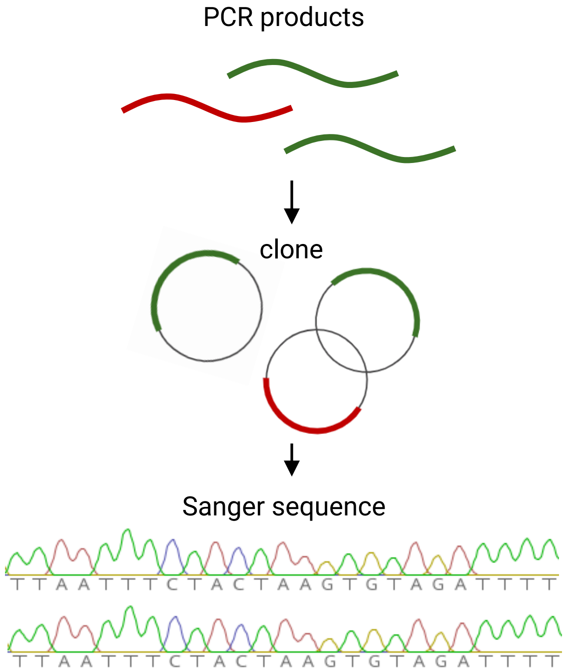


TIDE

ICE

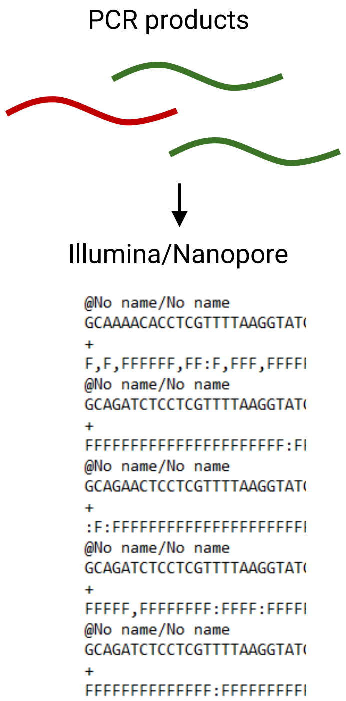
Sanger + deconvolution of trace files

Quick, easy, cheap
Can be used for populations (>10% mutations)



Clone + Sanger

More hassle



NGS

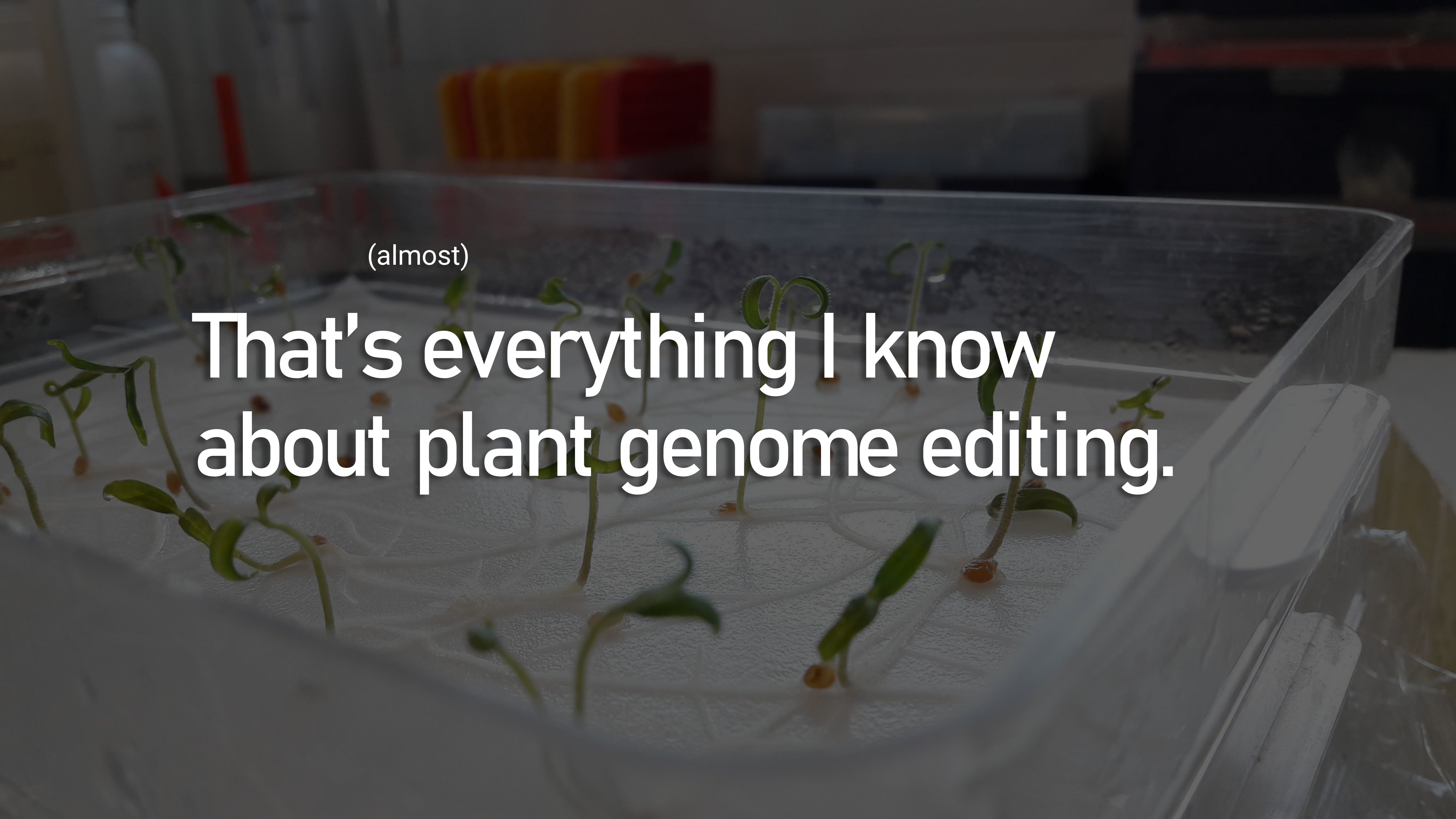
Can be great value, needs more
bioinformatics knowledge, sensitive

Method	Information	Sensitivity	Sample cost	TAT	Advantages	Limitations	When to use
Sanger sequencing and TIDE/ICE	Quantitative, DNA sequence	~10%	Sanger (+ cleanup)	2	-Easy -Inexpensive -Great value -Batch (ICE)	-Not all amplicons or mutations	-Stable mutant plants -Cell populations -Medium samples
Clone and sequence	Semi-quantitative, DNA sequence	How much $ you got?	Cloning + multiple Sangers (+ cleanup)	3	-No special equipment -Reliable -Multiplexing	-Slow -Hands-on time -Limited info.	-Few samples
Heteroduplex retention (PAGE)	Qualitative (can quantify), no DNA sequence	~10%	PCR and gel	1	-Fast -Can ID specific alleles	-Does not scale -Some amplicons don't work	Looking for specific mutant in various lines
T7 Endonuclease I or similar (heteroduplex)	Quantitative, no DNA sequence	~5-10%	$3.40 ~$2.50 enzyme	1	-Fast -Somewhat quantitative -Works for most amplicons	-Hands-on time -Some optimization required	-Cell populations -Homozygous yes/no? -Fast turnaround
qPCR/ddPCR	Quantitative, specific allele	0.01%-- depends	~6 euro	1	-Fast -quantitative -very sensitive	-Setup time -difficult to scale	-Have very specific repair product in mind AND it is rare in population
Illumina sequencing	Quantitative, DNA sequence	0.1-1%	Depends on scale… cents	7-21	-Can scale -Simple workflow -Power -Multiplexing	-Small amplicons (llumina) -Slow if using service	-Cell populations -A lot of samples
Nanopore	Quantitative, DNA sequence	~1%	30 euro for ~3000 reads, higher scales possible	1+	-Can scale -Simple workflow -Multiplexing -Longer amplicons than Illumina, can be several KB)	-Higher error rates than Illumina -Analysis methods not well developed	-Cell populations -Complex editing (e.g. promoter bashing) -Pre-testing Illumina amplicons (faster)

Nanopore: plasmidsaurus.com, check out their Premium PCR service. You can get amplicon sequencing for $15, but then you'll only receive reads that will align to your consensus. If you have multiple alleles, likely only one allele will turn out to be the consensus and the rest of the data is lost.

Genotyping pro tips from a postdoc who has been in the trenches

- ★ *PHIRE* Plant Direct PCR Kit
  - It's the most robust polymerase I've ever encountered.
  - It will amplify *anything* – be careful with your blanks!
  - https://www.thermofisher.com/order/catalog/product/F160L
- ★ Clear PCR bands can be sent for Sanger sequencing without purification
- ★ Can you use robotics at some step of the way?



(almost)

That's everything I know
about plant genome editing.

Tips

- Make sure your experimental design is sound
- Don’t get too hung up on gRNA design
- Become a pro at a modular cloning system
- Never underestimate genotyping
- Be curious about bioinformatics

★

- Have fun!

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